

Breast Cancer Diagnosis Prediction using K-Nearest Neighbors (KNN)

Author: Shafaat Ullah

Abstract

This project develops a K-Nearest Neighbors (KNN) classifier to predict whether a breast tumor is benign or malignant using the Breast Cancer Wisconsin Diagnostic dataset.

Introduction

Breast cancer is one of the most common cancers worldwide. Machine learning assists in early diagnosis by learning patterns from historical medical data.

Objectives

1. Analyze the dataset.
2. Perform EDA.
3. Preprocess the data.
4. Train a KNN model.
5. Evaluate performance.

Dataset

Records: 569

Features: 30 numeric features

Target: diagnosis (Benign/Malignant).

Software & Libraries

Python, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, Kaggle.

Exploratory Data Analysis

Reviewed dataset shape, columns, statistics, missing values, duplicates, target distribution, histograms, boxplots and correlation heatmap.

Preprocessing

Removed unnecessary empty column, encoded diagnosis (B=0, M=1), standardized features with StandardScaler, split data into 80% training and 20% testing.

Algorithm

K-Nearest Neighbors (KNN) classifies a sample based on the majority class among its K nearest training samples. Euclidean distance is commonly used.

Model Training

Model trained using KNeighborsClassifier with n_neighbors=5.

Evaluation

Training Samples: 455

Testing Samples: 114

Accuracy: 95.61%

Classification Report and Confusion Matrix generated.

Results

The model achieved high predictive performance and correctly classified most samples, making it suitable for educational demonstration.

Advantages

Simple, effective, non-parametric, performs well on standardized datasets.

Limitations

Sensitive to K value, scaling, and computational cost on large datasets.

Future Work

Tune K using Elbow Method, perform cross-validation, compare with SVM, Random Forest, and Logistic Regression.

Conclusion

The KNN model achieved 95.61% accuracy on the Breast Cancer Wisconsin dataset, demonstrating strong classification performance for early breast cancer diagnosis.

References

Scikit-learn Documentation; UCI Machine Learning Repository; Kaggle Breast Cancer Wisconsin Dataset.